

On the differential geometry of infinite-dimensional Lie groups and its applications to the hydrodynamics of perfect fluids*

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In the year 1765, L. Euler [8] published the equations of rigid body motion which bear his name. It does not seem useless to mark the 200th anniversary of Euler's equations by a modern exposition of the question.

The eulerian motions of a rigid body are the geodesics on the group of rotations of three dimensional euclidean space endowed with a left invariant metric. Basically, Euler's theory makes use of nothing but this circumstance; hence Euler's equations still hold for an arbitrary group. For the other groups, one obtains the "Euler equations" of rigid body motion in the n -dimensional space, the equations of the hydrodynamics of ideal fluids, etc.

Euler's theorem on the stability of the rotations around the longest and shortest axes of the inertia ellipsoid also has analogues in the case of an arbitrary group. In the case of hydrodynamics, this analogy is an extension of Rayleigh's theorem on the stability of flows whose velocity profile is inflexion free (see §10).

As another application of Euler's theory, we prove in §8, the explicit formula of the riemannian curvature of a group endowed with a left invariant metric. In §11, this formula is used in the study of the curvature of the group of diffeomorphisms, whose geodesics are ideal fluid flows.

In what follows, I tried, following the call of Bourbaki [6], to always substitute blind computations for Euler's lucid ideas.

1 Modern notations

Let G be a Lie group, $\mathfrak{L}$ its Lie algebra. A curve $g(t)$ is a mapping $g : \mathbb{R} \rightarrow G$. The velocity vector $\dot{g} = dg/dt$ belongs to the tangent space to G at the point g ; we shall denote this tangent space TG_g . Obviously, $TG_e = \mathfrak{L}$.

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¹ Most of the results have been announced in [1, 2, 3, 4].

Let $g \in G$. The left and right translations L_g and R_g are mappings from the group G into G , defined by

$$L_g h = gh, \quad R_g h = hg, \quad (h \in G). \quad (1)$$

The mappings induced on the tangent spaces will be denoted

$$\tilde{L}_g : TG_h \rightarrow TG_{gh}, \quad \tilde{R}_g : TG_h \rightarrow TG_{hg}. \quad (2)$$

We shall denote Ad_g the mapping from the algebra $\mathfrak{U}$ to itself

$$Ad_g \xi = \tilde{R}_{g^{-1}} L_g \xi, \quad (\xi \in \mathfrak{U}). \quad (3)$$

We shall denote $\exp : \mathfrak{U} \rightarrow G$ the natural map from the algebra into the group;

$$\exp(ft) = g(t), \quad (t \in \mathbb{R}, f \in \mathfrak{U}, g \in G) \quad (4)$$

is a one parameter group whose velocity vector is $dg/dt|_{t=0} = f$. We shall denote, for $\xi \in \mathfrak{U}, \eta \in \mathfrak{U}$

$$[\xi, \eta] \in \mathfrak{U}$$

the algebra commutation operation, which is bilinear and defined by

$$\exp[\xi, \eta] = \exp \xi \exp \eta \exp(-\xi) \exp(-\eta) + O(\xi^2) + O(\eta^2).$$

From the definition of the commutator and from (3) it follows that:

$$Ad_{\exp(ft)} \xi = \xi + t[f, \xi] + O(t^2), \quad (t \rightarrow 0). \quad (5)$$

The commutator is antisymmetric and satisfies the Jacobi identity:

$$[\xi + t\eta, \zeta] = [\xi, \eta] + t[\eta, \zeta], \quad [\xi, \eta] = -[\eta, \xi]; \quad (\xi, \eta, \zeta \in \mathfrak{U}),$$

$$[[\xi, \eta], \zeta] + [[\eta, \zeta], \xi] + [[\zeta, \xi], \eta] = 0 \quad (6)$$

The operator Ad_g is an algebra automorphism; for g variable, the operators Ad_g form an antirepresentation of the group:

$$Ad_g[\xi, \eta] = [Ad_g \xi, Ad_g \eta]; \quad Ad_{gh} = Ad_h Ad_g. \quad (7)$$

The dual of the tangent space TG_g will be called the cotangent space and denoted T^*G_g . An element $\xi \in T^*G_g$ is a linear form on TG_g ; its value on η , will be denoted

$$(\xi, \eta) \in \mathbb{R}, \quad \xi \in T^*G_g, \quad \eta \in TG_g.$$

The conjugate operators of $\tilde{L}_g, \tilde{R}_g$ will be denoted

$$\tilde{L}_g^* : T^*G_{gh} \rightarrow T^*G_h, \quad \tilde{R}_g^* : T^*G_{hg} \rightarrow T^*G_h.$$

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These operators are defined by

$$(\tilde{L}_g^* \xi, \eta) = (\xi, \tilde{L}_g \eta), \quad (\tilde{R}_g^* \xi, \eta) = (\xi, \tilde{R}_g \eta).$$

Finally, the operator $Ad_g^* : \mathfrak{U}^* \rightarrow \mathfrak{U}^*$ is defined by

$$(Ad_g^* \xi, \eta) = (\xi, Ad_g \eta). \quad (8)$$

Let $\mathfrak{L} : \mathfrak{U}^* \rightarrow \mathfrak{U}^*$ be a symmetric operator:

$$(\mathfrak{L} \xi, \eta) = (\mathfrak{L} \eta, \xi). \quad \xi, \eta \in \mathfrak{U}.$$

Let us define the symmetric operator $\mathfrak{L}_g : TG_g \rightarrow T^*G_g$ by left translation:

$$\mathfrak{L}_g \xi = \tilde{L}_{g^{-1}}^* \mathfrak{L} \tilde{L}_{g^{-1}} \xi.$$

The operator $\mathfrak{L}$ will be supposed to be positive definite, which means that the scalar product of $\xi, \eta \in TG_g$

$$\langle \xi, \eta \rangle_g = (\mathfrak{L}_g \xi, \eta) = (\mathfrak{L}_g \eta, \xi) = \langle \eta, \xi \rangle_g \quad (9)$$

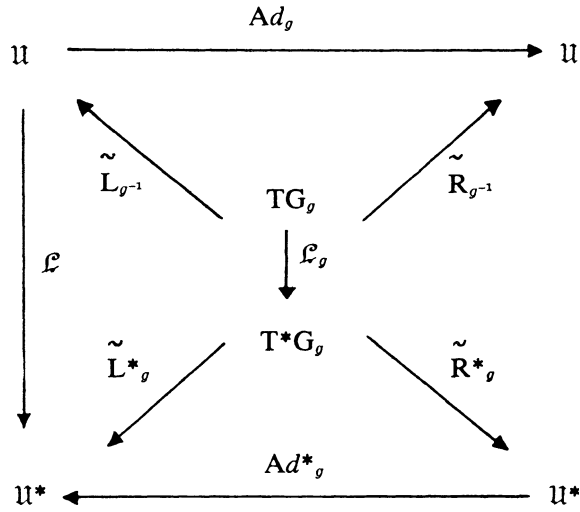
is a positive definite form. The product $\langle, \rangle_g$ defines on G a left invariant riemannian metric.

The product $\langle, \rangle_g$ in the algebra will be denoted $\langle, \rangle$. Let us introduce the operation $B : \mathfrak{U} \rightarrow \mathfrak{U}$ defined by

$$\boxed{\langle [a, b], c \rangle = \langle B(c, a), b \rangle \quad \text{for all } b \in \mathfrak{U}.} \quad (10)$$

Of course, $B(c, a)$ depends bilinearly on c and a and, for fixed c , $\langle B(c, a), b \rangle$ is antisymmetric in a and b :

$$\langle B(c, a), b \rangle + \langle B(c, b), a \rangle = 0 \quad (11)$$



2 Classical notations

In rigid body mechanics, the objects defined above have the following names:

$G = SO(3)$	the configuration space;
g	rigid body configuration;
$g(t)$	rigid body motion;
$\dot{g}$	rotational velocity;
$\mathfrak{U} = \mathbb{R}^3$	space of rotation vectors;
$[\xi, \eta]$	vector product in $\mathbb{R}^3$;
$\omega_c = \tilde{L}_{g^{-1}} \dot{g} \in \mathfrak{U}$	rotation vector w.r.t. the body;
$\omega_s = \tilde{R}_{g^{-1}} \dot{g} = Ad_g \omega_c$	rotation vector w.r.t. space;
$\mathcal{L}_g$	inertia operator;
$M = \mathcal{L}_g \dot{g}$	angular momentum;
$\mathfrak{U}^*$	space of angular momenta;
$M_c = \tilde{L}_g^* M = \mathcal{L} \omega_c$	angular momentum w.r.t. the body;
$M_s = \tilde{R}_g^* M = Ad_{g^{-1}}^* M_c$	angular momentum w.r.t. space;
$T = \frac{1}{2} \langle \dot{g}, \dot{g} \rangle_g = \frac{1}{2} \langle \omega_c, \omega_c \rangle = \frac{1}{2} \langle M_c, \omega_c \rangle = \frac{1}{2} \langle M, \dot{g} \rangle$	the kinetic energy.

The principle of least action asserts that the motions of a rigid body around a fixed point are (in the absence of exterior forces) the geodesics² of the group G endowed with a left invariant metric (9). To each geodesic $g(t)$ correspond 4 curves

$$\omega_c(t) \in \mathfrak{U}, \quad \omega_s(t) \in \mathfrak{U}, \quad M_c(t) \in \mathfrak{U}^*, \quad M_s(t) \in \mathfrak{U}^*. \quad (12)$$

3 Euler's theorems

For the curves (12) and $G = SO(3)$, Euler proved the following two theorems

Theorem 1. *The curves (12) satisfy the Euler equation*

$$\boxed{\frac{dM_c}{dt} = [M_c, \omega_c]} \quad (13)$$

Theorem 2. *The angular momentum with respect to space is invariant:*

$$\boxed{\frac{dM_s}{dt} = 0.} \quad (14)$$

With the notations of §1, one can write formula (13) in the form

$$\boxed{\frac{d\omega_c}{dt} = B(\omega_c, \omega_c)} \quad \omega_c = \tilde{L}_{g^{-1}} \dot{g}. \quad (15)$$

We shall prove formulæ (14) and (15) for the case of an arbitrary Lie group.

² A curve $g(t)$ in the riemannian space is a geodesic, if it is an extremal of the “action” integral $\frac{1}{2} \int \langle \dot{g}, \dot{g} \rangle dt$

4 Proof of Euler's first theorem

$$\frac{d\omega_c}{dt} = F(\omega_c).$$

In particular, the mapping $\exp : \mathfrak{U} \rightarrow G$ allows us to consider the neighborhood of 0 in $\mathfrak{U}$ as a chart of the neighborhood of e in G . The tangent space to $\mathfrak{U}$ at a , $T\mathfrak{U}_a$, is naturally identified with $\mathfrak{U}$.

Lemma 1. *Let us consider the operator $\tilde{L}_a$, induced by the left translation $L_{\exp a}$:*

$$\tilde{L}_a : \mathfrak{U} = T\mathfrak{U}_0 \rightarrow T\mathfrak{U}_a = \mathfrak{U}. \quad (16)$$

For this operator

$$\boxed{\tilde{L}_a \xi = \xi + \frac{1}{2}[a, \xi] + O(a^2)} \quad |a| \rightarrow 0. \quad (17)$$

Proof of lemma 1. It is known that all computations on Lie groups and Lie algebras may be done as if on a matrix group. For example, for any $a, b \in \mathfrak{U}$

$$(\exp a)(\exp b) = \exp \left(a + b + \frac{1}{2}[a, b] + O(a^2) + O(b^2) \right) \quad (18)$$

$$(a, b \rightarrow 0).$$

Setting $b = \xi t, t \rightarrow 0, |a| \rightarrow 0$ in (18) one finds

$$(\exp a)(\exp \xi t) = \exp \left[a + \left(\xi + \frac{1}{2}[a, \xi] + O(a^2) \right) t + O(t^2) \right], \quad (t \rightarrow 0)$$

which is equivalent to (17). Lemma 1 is proved.

Proof of formula (15). Now, let q be cartesian coordinates in the algebra $\mathfrak{U}$, endowed with a metric $\langle \cdot, \cdot \rangle$. In a neighborhood of $e \in G$, one can use q as local coordinates. Let $g(t)$ be a curve in G . According to lemma 1, the coordinates $\dot{q} \in \mathfrak{U}$ of the velocity vector $\dot{g}$ are related to the vector³ $\omega = \tilde{L}_{g^{-1}} \dot{g}$ by the formula

$$\omega = \dot{q} - \frac{1}{2}[q, \dot{q}] + O(q^2) \quad |q| \rightarrow 0. \quad (19)$$

Hence, the Lagrange function $L(q, \dot{q})$ is given by the formula

$$2L = \langle \omega, \omega \rangle = \langle \dot{q}, \dot{q} \rangle - \langle \dot{q}, [q, \dot{q}] \rangle + O(q^2), \quad (|q| \rightarrow 0) \quad (20)$$

Using (10), (19) in order to compute the partial derivatives of L , one finds the impulse $p = \partial L / \partial \dot{q}$:

$$\begin{aligned} p &= \dot{q} - \frac{1}{2}[q, \dot{q}] - \frac{1}{2}B(\dot{q}, q) + O(q^2) \\ &= \omega - \frac{1}{2}B(\omega, q) + O(q^2). \end{aligned} \quad (21)$$

According to (19), (20), (21), the Euler-Lagrange equation

$$\dot{p} = \frac{\partial L}{\partial q},$$

³ Here and in the following computation, the index c is omitted in order to simplify the formulæ.

that is

$$\dot{p} = \frac{1}{2}B(\dot{q}, \dot{q}) + O(q) = \frac{1}{2}B(\omega, \omega) + O(q) \quad (22)$$

becomes

$$\dot{\omega} - \frac{1}{2}B(\omega, \omega) = \frac{1}{2}B(\omega, \omega) + O(q). \quad (23)$$

When $q = 0$, (23) becomes (15). Theorem 1 is proved.

Remark. It follows also from (21), (22) that for a geodesic $q(t)$ with $q(0) = 0$, one has

$$\ddot{q}(0) = B(\dot{q}(0), \dot{q}(0)) = B(\omega, \omega). \quad (24)$$

5 Proof of Euler's second theorem

Euler's second theorem is a special case of "Noether's theorem". But, for what follows, it is useful to compute independently dM_s/dt . Here, in accordance with §2, the vector $M_s \in \mathfrak{U}^*$ is

$$M_s(t) = Ad_{g^{-1}(t)} \mathfrak{L}\omega_c(t).$$

Lemma 2. *Let $g(t)$ be a curve in the group G and ζ be an element of the algebra $\mathfrak{U}$. In the algebra, let us consider the curve defined by*

$$A(t) = Ad_{g^{-1}(t)} \zeta \quad (25)$$

Then

$$\frac{d}{dt}A(t) = [A(t), \xi(t)] \quad (26)$$

where

$$\xi(t) = \tilde{L}_{g^{-1}} \frac{dg}{dt}. \quad (27)$$

Proof of lemma 2. According to (27) one has

$$g(t+dt) = g(t) \exp(\xi(t)dt + O(dt^2)).$$

Hence, substituting (25) in (3) and neglecting $O(dt^2)$, one finds

$$\begin{aligned} A(t+dt) &= A_{g^{-1}(t+dt)} \zeta = \tilde{R}_{\exp(\xi dt)} \tilde{L}_{\exp(-\xi dt)} \tilde{R}_g \tilde{L}_{g^{-1}} \zeta = \\ &= \tilde{R}_{\exp(\xi dt)} \tilde{L}_{\exp(-\xi dt)} A(t) = Ad_{\exp(-\xi dt)} A(t). \end{aligned}$$

According to (5), this last expression is

$$Ad_{\exp(-\xi dt)} A(t) = A(t) + [A, \xi]dt + O(dt^2)$$

which proves lemma 2.

Lemma 3. *If, in lemma 2, $g(t) = e^{ft}$, then*

$$A(t) = \zeta + t[\zeta, f] + \frac{t^2}{2}[[\zeta, f], f] + \dots \quad (28)$$

Proof of lemma 3. The vector (27), $\xi(t) = f$ does not depend on t for $g(t) = e^{ft}$. Hence (26) gives

$$\frac{dA}{dt} = [A, f], \quad \frac{d^2A}{dt^2} = [[A, f], f], \dots$$

But as $A(0) = \zeta$, (28) is proved.

Remark. If $A(t) = Ad_{g(t)}\zeta$, then

$$\frac{d}{dt}A(t) = [A(t), \eta(t)], \quad (29)$$

where

$$\eta(t) = -\tilde{R}_{g^{-1}} \frac{dg}{dt}. \quad (30)$$

Because, in accordance with lemma 2 (29) holds with

$$\eta(t) = \tilde{L}_g \frac{d}{dt}g^{-1}(t), \quad \text{et} \quad \frac{d}{dt}g^{-1}(t) = -\tilde{R}_{g^{-1}}\tilde{L}_{g^{-1}} \frac{dg}{dt}.$$

Proof of formula (14). By definition,

$$M_s(t) = Ad_{g^{-1}(t)}^* \mathfrak{L}\omega_c(t) \in \mathfrak{U}^*. \quad (31)$$

Let us consider $\zeta \in \mathfrak{U}$. According to (8), (31), (9),

$$(M_s, \zeta) = (\mathfrak{L}\omega_c, Ad_{g^{-1}}\zeta) = \langle \omega_c, Ad_{g^{-1}}\zeta \rangle.$$

Now, theorem 1 and lemma 2 give

$$\frac{d(M_s, \zeta)}{dt} = \langle B(\omega_c, \omega_c), A \rangle + \langle \omega_c, [A, \omega_c] \rangle$$

with $A = Ad_{g^{-1}}\zeta$. Hence, by definition of the operation B (10),

$$d(M_s, \zeta)/dt = 0$$

for all ζ , which proves Euler's second theorem.

6 Stationary motions

Now, let us consider more carefully the system of Euler's equations

$$\boxed{\frac{d\omega}{dt} = B(\omega, \omega)} \quad (32)$$

which determine the evolution of the vector ω in the algebra $\mathfrak{U}$. Let v be a stationary point of system (32):

$$\boxed{B(v, v) = 0}. \quad (33)$$

Then, *the geodesic $g(t)$, originating from the point e with initial velocity $\dot{g}(0) = v$, is a subgroup: $g(t) = \exp vt$* . Because according to (15), (33), for all t :

$$\tilde{L}_{g^{-1}} \frac{dg}{dt} = v = \tilde{L}_{\exp(-vt)} \frac{d\exp(vt)}{dt}.$$

Definition 1. The geodesic $g(t)$ is called a “stationary motion”, if it is a one parameter subgroup.

We just proved that *the stationary motions are the only motions for which the “rotation vector with respect to the body” $\omega = \tilde{L}_g^{-1} \dot{g}$ is constant: $\dot{\omega} = 0$* .

For the classical rigid body ($G = SO(3)$) the stationary motions are the rotations around the great, the middle, or the small axis of the “inertia ellipsoid”. In the general case, the stationary motions possess an interesting property of extremality. In order to describe this property, we start with some remarks.

Lemma 4. *The system (32) admits the energy first integral*

$$E = \frac{1}{2} \langle \omega, \omega \rangle. \quad (34)$$

Proof. According to (32), (10)

$$\dot{E} = \langle \omega, \dot{\omega} \rangle = \langle \omega, B(\omega, \omega) \rangle = \langle [\omega, \omega], \omega \rangle = 0, \quad \text{C.Q.F.D.}$$

Definition 2. Under the name “foliation”, I shall include every decomposition into equivalence classes of an arbitrary equivalence relation. I shall say that a foliation of the algebra $\mathfrak{U}$ is invariant, if with a point $\omega(0)$, the whole trajectory $\omega(t)$ of the system (32) belongs to one and the same leaf.

Example. According to lemma 4, the decomposition of $\mathfrak{U}$ into level surfaces of E is an invariant foliation. The leaves are the spheres $\langle \omega, \omega \rangle = 2E > 0$ and the point $\omega = 0$.

Euler's second theorem provides another example of an invariant foliation.

Definition 3. The vectors $\omega \in \mathfrak{U}$ and $\omega' \in \mathfrak{U}$ are “isorotating”,

$$\omega \sim \omega'$$

if there exists $g \in G$ such that

$$Ad_g^* \mathfrak{L}\omega = \mathfrak{L}\omega'.^4 \quad (35)$$

Lemma 5. *The relation $\sim$ defines an invariant foliation in $\mathfrak{U}$.*

Proof. Obviously, $\omega \sim \omega$. Equation (7) implies the relations $(\omega \sim \omega') \Rightarrow (\omega' \sim \omega)$ and $(\omega \sim \omega' \sim \omega'') \Rightarrow (\omega \sim \omega'')$. Hence $\sim$ is an equivalence relation. The invariance of the leaves results from (14), (31), (35). Lemma 5 is proved.

In what follows, the words leaf, foliation, will indicate the foliation of the algebra $\mathfrak{U}$ into leaves of isorotating ω . Obviously, these leaves are the linear images of orbits of the adjoint action Ad_ω of the group in the algebra.⁵

Examples. For $G = SO(3)$ (rigid body) the leaves are concentric ellipsoids and the point

$$\langle \langle M_s(\omega), M_s(\omega) \rangle \rangle = C$$

where $\langle \langle, \rangle \rangle$ is the bi-invariant product in $\mathfrak{U}^*$. For $G = SO(n)$, the leaves are the “flag manifolds”. For the group of homotheties and translations of a line, the leaves are two half-planes and all the points of the line which separates them.

Theorem 3. *The stationary points v of the system (32) are the conditional extrema ω of the energy E on the leaves made of the ω 's isorotating with v .*

For the proof of theorem 3, we shall use some computations.

Lemma 6. *Let $\omega \in \mathfrak{U}, f \in \mathfrak{U}$, and*

$$\omega(t) = \mathfrak{L}^{-1} Ad_{e^{ft}}^* \mathfrak{L}\omega \quad (36)$$

Then :

$$\frac{d\omega}{dt} = B(\omega, f), \quad (37)$$

$$\omega(t) = \omega + tB(\omega, f) + \frac{t^2}{2}B(B(\omega, f), f) + \dots \quad (38)$$

Proof. For any $\zeta \in \mathfrak{U}$, using (36), (9), (8), (9), one finds :

$$\begin{aligned} \langle \omega(t), \zeta \rangle &= \langle \mathfrak{L}^{-1} Ad_{e^{ft}}^* \mathfrak{L}\omega, \zeta \rangle = \langle Ad_{e^{ft}}^* \mathfrak{L}\omega, \zeta \rangle = \\ &= \langle \mathfrak{L}\omega, Ad_{e^{ft}} \zeta \rangle = \langle \omega, Ad_{e^{ft}} \zeta \rangle. \end{aligned} \quad (39)$$

For every $\zeta \in \mathfrak{U}$, according to (39), (29) and (10), one has for $t = 0$

$$\langle \frac{d\omega}{dt}, \zeta \rangle = \langle \omega, \frac{dA}{dt} \rangle = \langle \omega, [f, A] \rangle = \langle B(\omega, f), \zeta \rangle, \quad (40)$$

⁴ In rigid body dynamics, the isorotating ω correspond to the same angular momentum. In hydrodynamics, they are the “isorotational” velocity fields (see[1]).

⁵ transl. note: Nowadays these ‘isorotating leaves’ are usually viewed in the dual Lie algebra where they form the famous co-adjoint orbits

where $A(t) = Ad_{e^{ft}} \zeta$. As ζ is arbitrary, formula (40) proves (37) for $t = 0$. Formula (36) implies the identity

$$\omega(t_1 + t_2, \omega(0)) = \omega(t_1, \omega(t_2, \omega(0))).$$

Hence, formula (37) is proved for all t . Hence the derivatives

$$\frac{d^2 \omega}{dt^2} = B(B(\omega, f), f); \dots; \frac{d^n \omega}{dt^n} = B(\dots (B(\omega, f), \dots, f); \dots$$

which is equivalent to (38). Lemma 6 is proved.

Proof of theorem 3. Let ξ be a vector tangent to a leaf at the point v . As the leaves are orbits of a differentiable action of the group G (see (35)), there exists $f(\xi) \in \mathfrak{U}$ such that

$$v + \xi t = \omega(t) + O(t^2) \quad t \rightarrow 0$$

where the function $\omega(t)$ is defined by (36) with $\omega(0) = v$.

According to (37) one has

$$\xi = B(v, f). \quad (41)$$

In accordance with (34), (41), (11), the total differential of the energy on the leaf is

$$dE = \langle v, \xi \rangle = \langle v, B(v, f) \rangle = -\langle B(v, v), f \rangle. \quad (42)$$

Formula (42) shows that $dE = 0$ for all ξ if and only if $B(v, v) = 0$. C.q.f.d., according to (33).

7 Stability of stationary motions

Consider a system of ordinary differential equations

$$\dot{x} = f(x). \quad x \in \mathbb{R}^n. \quad (43)$$

Definition 4. A point x_0 (with $f(x_0) = 0$) is stable if

$$\forall \varepsilon > 0, \exists \delta > 0 : (|x(0) - x_0| < \delta) \Rightarrow (|x(t) - x_0| < \varepsilon \quad \forall t > 0).$$

Suppose now that a foliation invariant with respect to the system (43) is defined in the space $\mathbb{R}^n$. The point x_0 is a *regular point*, if there exists, on a neighborhood of x_0 , a system of differentiable local coordinates y ($y(x_0) = 0$) such that the connected components of the intersections of the leaves with the neighborhood are parallel planes of a fixed dimension k for the coordinates y .

Let E be a first integral of system (43). Suppose that

- 1) x_0 is a conditional extremum point of E on the leaf;
- 2) x_0 is a regular point and that
- 3) the second derivative of E on the leaf at x_0 is a non degenerate quadratic form.

One proves easily a lemma well known since Lagrange:

Lemma 7. *A point x_0 satisfying conditions 1), 2), 3) is a stationary point of system (43). If, moreover, the second derivative of E on the leaf is a positive-definite (or negative-definite) quadratic form, the stationary point x_0 is stable.*

Indeed, in coordinates y ,

$$\begin{aligned} E(y) &= \frac{1}{2}(E_2y, y) + O(y^3) \quad \text{ou} \quad (E_2y, z) = (y, E_2z), \\ \dot{E} &= (E_2y, \dot{y}) + O(y^2)\dot{y}. \quad (y \rightarrow 0). \end{aligned}$$

If $\dot{y} \neq 0$ for $y = 0$, then for y small enough $(E_2y, y) \neq 0$ and E is not a first integral. Hence $\dot{y} = 0$ and x_0 is a stationary point. The claim of stability is obvious (Lagrange, Dirichlet, etc.).

Let us apply lemma 7 to system (32).

Let v be a stationary point of Euler's system (32). Suppose that v is a regular point, i.e. that v belongs to a leaf of maximal dimension. Consider the quadratic form

$$\boxed{2d^2E = \langle B(v, f), B(v, f) \rangle + \langle [f, v], B(v, f) \rangle.} \quad (44)$$

Theorem 4. *If the quadratic form (44) is positive-definite or negative-definite, the stationary point v is stable.*

Here the form d^2E is considered as a quadratic form on the vector space tangent to the leaf

$$\xi = B(v, f), \quad f \in \mathfrak{L}.$$

Proof of theorem 4. According to lemma 7, it is enough to prove that the second derivative of the energy on the leaf has the form (44). Formula (38) defines an algebra mapping on the leaf, which transforms

$$\begin{aligned} f &\rightarrow \omega = v + \xi_1 + \xi_2 + O(f^3), \\ \xi_1 &= B(v, f), \quad 2\xi_2 = B(B(v, f)). \end{aligned}$$

Substituting ω in (34) and taking (42), (33) into account, one finds

$$2d^2E = \langle \xi_1, \xi_1 \rangle + 2\langle$$

which is equivalent to (44).

It follows from the definition of d^2E that the form (44) does not depend on f , but rather only on $\xi_1 = B(v, f)$. In fact, one can check this by a small computation:

Lemma 8. *The bilinear form $C(x, y) = \langle [x, v], B(v, y) \rangle$ is symmetric:*

$$C(x, y) = C(y, x). \quad (45)$$

Proof. According to (11), (10), (6), (10), (33)

$$\begin{aligned}
 \langle [x, v], B(v, y) \rangle &= \langle B(v, [v, x]), y \rangle = \langle [[v, x], y], v \rangle \\
 &= \langle [v, [x, y]], v \rangle + \langle [x, [y, v]], v \rangle = \\
 &= \langle B(v, v), [x, y] \rangle + \langle B(v, x), [y, v] \rangle = \\
 &= \langle [y, v], B(v, x) \rangle. \quad C.Q.F.D.
 \end{aligned}$$

Let now $B(v, f_1) = B(v, f_2)$. Let us show that the corresponding values of d^2E coincide. We set $x = f_1 - f_2$, $y = f_1$. Then $B(v, x) = 0$. According to (44), (45)

$$\begin{aligned}
 2(d^2E(f_1) - d^2E(f_2)) &= \langle [x, v], B(v, y) \rangle = \\
 &= \langle [y, v], B(v, x) \rangle = 0.
 \end{aligned}$$

Hence the quadratic form (44) depends only on $\xi = B(v, f)$. Theorem 4 now follows immediately from lemma 7.

Remark. Consider the “variational equations” of Euler’s equations (32) in the neighborhood of v :

$$\dot{\xi} = B(v, \xi) + B(\xi, v). \quad (46)$$

As the leaves and the energy are invariant, the form (44) must be a first integral of system (46). Actually, one can check the conservation of d^2E by a direct computation.

Indeed, according to (44), (45), (41)

$$\frac{d}{dt}d^2E = \langle \xi, \dot{\xi} \rangle + \langle [f, v], \dot{\xi} \rangle.$$

Substituting $\dot{\xi}$ from (46), one gets by using successively (11), (45), (41), (10), (45), (11):

$$\begin{aligned}
 \frac{d}{dt}d^2E &= \\
 &\langle \xi, B(v, \xi) \rangle + \langle \xi, B(\xi, v) \rangle + \langle [f, v], B(v, \xi) \rangle + \langle [f, v], B(\xi, v) \rangle = \\
 &\langle \xi, B(\xi, v) \rangle + \langle [\xi, v], \xi \rangle + \langle [v, [f, v]], \xi \rangle = \\
 &= \langle [v, [f, v]], B(v, f) \rangle = -\langle [f, v], B(v, [f, v]) \rangle = 0, \quad C.Q.F.D.
 \end{aligned}$$

8 The riemannian curvature of Lie groups

As an example of application of Euler’s theorems, I shall compute the connexion (55) and the riemannian curvature (53) of a Lie group, endowed with a left invariant metric (compare to [3]).

Let M be a riemannian manifold, $x \in M$ and $\xi \in TM_x$ a tangent vector to M at x . We shall denote

$$\gamma(x, \xi, t) = \gamma(\xi, t) = \gamma(t) = \gamma$$

On the differential geometry of infinite dimensional Lie groups

the geodesic⁶ on M , originating from $x = \gamma(0)$ with initial velocity $\xi = \dot{\gamma}(0)$.

One determines the *parallel transport along γ*

$$\pi_{\gamma(t)} : TM_{\gamma(0)} \rightarrow TM_{\gamma(t)}$$

with the help of the following construction.

Let

$$P_{\gamma(\xi, t)} \eta = \frac{1}{t} \frac{d}{d\tau} \Big|_{\tau=0} \gamma(x, \xi + \eta \tau, t) \in TM_{\gamma(t)}. \quad (47)$$

Then

$$\pi_{\gamma(t)} \eta = P_{\gamma(t)} \eta + O(t) \quad (t \rightarrow 0).$$

The *covariant derivative* $\nabla_\xi \eta$ of a tangent vector-field η along the direction of the tangent vector $\xi \in TM_x$ is, by definition, the vector-field

$$\nabla_\xi \eta = \frac{d}{dt} \Big|_{t=0} P_{\gamma(t)}^{-1} \eta(\gamma(\xi, t)) = \frac{d}{dt} \Big|_{t=0} \pi_{\gamma(t)}^{-1} \eta(\gamma(\xi, t)) \in TM_x. \quad (48)$$

Let ξ, η be two orthogonal unit vectors in TM_x . One calls⁷ the *riemannian sectional curvature* of M along ξ, η the real number

$$C_{\xi\eta} = \langle -\nabla_\xi \nabla_\eta \xi, \eta \rangle + \langle \nabla_\eta \nabla_\xi \xi, \eta \rangle - \langle \nabla_{\{\xi, \eta\}} \xi, \eta \rangle, \quad (49)$$

where $\langle \cdot, \cdot \rangle$ is the scalar product determining the metric, and ξ, η denote two vector-fields. Finally, the vector-field $\{\xi, \eta\}$ denotes the *Poisson bracket*⁸ of the two vector-fields ξ, η .

With regard to Poisson bracket, we shall use the following known things. To each vector-field ζ on M , corresponds a one-parameter group of diffeomorphisms of M

$$\exp \zeta t : x \mapsto z(x, t) \in M; \quad z(x, 0) = x, \quad \frac{dz}{dt} = \zeta.$$

The *Poisson bracket of the vector-fields ξ, η tangent to M is defined as the commutator in the Lie algebra*⁹: the vector-field $\{\xi, \eta\}$ depends bilinearly on ξ, η and satisfies the equation:

$$\exp \zeta t \exp \eta s \exp(-\xi t) \exp(-\eta s) = \exp\{\xi, \eta\} st + O(s^2) + O(t^2). \quad (50)$$

In a system of local coordinates x on M the Poisson bracket is given by the formula:

$$\{\xi, \eta\} = \frac{\partial \xi}{\partial x} \eta - \frac{\partial \eta}{\partial x} \xi. \quad (51)$$

⁶ That is the extremal of the “least action principle” $\delta \int \dot{\gamma}^2 dt = 0$.

⁷ See for example [13].

⁸ transl. note: Nowadays called the *Lie bracket*

⁹ And differs by the sign from [13].

Indeed, computing the left hand side of (50), one finds

$$\begin{aligned} \exp \xi t : x &\mapsto x + \xi t + O(t^2), \\ \exp \xi t \exp \eta s : x &\mapsto x + \xi t + \eta s + \frac{\partial \xi}{\partial x} \eta s t + O(s^2) + O(t^2), \\ \exp \xi t \exp \eta s \exp(-\xi t) \exp(-\eta s) : \\ x &\mapsto x + \left(\frac{\partial \xi}{\partial x} \eta - \frac{\partial \eta}{\partial x} \xi \right) s t + O(s^2) + O(t^2), \end{aligned}$$

which proves formula (51).

Let now M be a Lie group G , $\mathfrak{U}$ its Lie algebra, ξ and η two *right*-invariant vector-fields. Then the vector-field $\{\xi, \eta\}$ also is right-invariant. A right-invariant vector-field is well-defined by its value at the neutral element, $\xi_e \in \mathfrak{U}$. *The Poisson bracket of the right-invariant vector-fields coincide at the point e with the commutator of the Lie algebra:*

$$\{\xi, \eta\}_e = [\xi_e, \eta_e].$$

Indeed, let $\exp(\xi t)$ be a one-parameter group of diffeomorphisms of the group G , corresponding to the right-invariant vector-field ξ . Then, for $x \in G$

$$\exp(\xi t) : x \mapsto (\exp \xi_e t)x$$

where $\exp \xi_e t$ is the one-parameter subgroup of the group G . Hence

$$\exp \xi t \exp \eta s \exp(-\xi t) \exp(-\eta s) : x \mapsto \exp \xi_e t \exp \eta_e s \exp(-\xi_e t) \exp(-\eta_e s)x.$$

C.Q.F.D.

Finally, let ξ and η be two *left*-invariant vector-fields on G . *The value at e of the Poisson bracket of left-invariant vector-fields differs by the sign of the commutator in the Lie algebra :*

$$\{\xi, \eta\}_e = -[\xi_e, \eta_e]. \quad (52)$$

Indeed, a left-invariant vector-field on G is a right-invariant vector-field on the group G' with the operation $g_1 \star g_2 = g_2 g_1$. But the commutator of the Lie algebra $\mathfrak{U}'$ of the group G' differs by the sign from the commutator in the algebra $\mathfrak{U}$.

Now, let as above G be a Lie group, endowed with a left-invariant metric, $\mathfrak{U}$ its Lie algebra, $\langle, \rangle$ the scalar product in the algebra, which determines the metric, $[\cdot, \cdot]$ the Lie bracket in the algebra, and B the operation defined in §1.

Theorem 5. *The sectional curvature of the group G at the point e for the plane defined by the two orthogonal and unitary vectors $\xi, \eta \in \mathfrak{U}$, is given by the formula*

$$\boxed{C_{\xi\eta} = \langle \delta, \delta \rangle + 2\langle \alpha, \beta \rangle - 3\langle \alpha, \alpha \rangle - 4\langle B_\xi, B_\eta \rangle} \quad (53)$$

where

$$\left. \begin{aligned} 2\delta &= B(\xi, \eta) + B(\eta, \xi), & 2\beta &= B(\xi, \eta) - B(\eta, \xi), \\ 2\alpha &= [\xi, \eta], & 2B_\xi &= B(\xi, \xi), & 2B_\eta &= B(\eta, \eta). \end{aligned} \right\} \quad (54)$$

Remark. For a bi-invariant metric, one has $B(\xi, \eta) = [\xi, \eta]$, hence $B_\xi = B_\eta = \delta = 0$, $\beta = 2\alpha$. Thus, (53) becomes the known formula

$$C_{\xi\eta} = \frac{1}{4} \langle [\xi, \eta], [\xi, \eta] \rangle.$$

In order to prove theorem 5, we shall at first express the covariant derivative in terms of B operations.

Lemma 9. *Let ξ, η be two left-invariant vector-fields on the group G . Then at the point e , the left-invariant vector-field $\nabla_\xi \eta$ is given by the formula*

$$\boxed{\nabla_\xi \eta = \frac{1}{2}([\xi, \eta] - B(\xi, \eta) - B(\eta, \xi)).} \quad (55)$$

Proof of lemma 9. One first computes the parallel transport (47) in algebra coordinates (§4). According to (24), the geodesics of the group have the following expression in the algebra

$$\gamma(0, \xi, t) = t\xi + \frac{t^2}{2}B(\xi, \xi) + O(t^3) \quad t \rightarrow 0. \quad (56)$$

It follows from (56) that the vector

$$P_{\gamma(\xi, t)} \eta = \frac{1}{t} \frac{d}{d\tau} \Big|_{\tau=0} \gamma(0, \xi + \eta\tau, t) \in T\mathfrak{U}_\gamma = \mathfrak{U}$$

has coordinates

$$P_{\gamma(\xi, t)} \eta = \eta + \frac{t}{2}(B(\xi, \eta) + B(\eta, \xi)) + O(t^2). \quad (57)$$

By definition (48), for a left-invariant vector-field η

$$\nabla_\xi \eta = \frac{d}{dt} \Big|_{t=0} P_{\gamma^{-1}(\xi, t)} \tilde{L}_{\gamma(\xi, t)} \eta. \quad (58)$$

Substituting in (58) the expressions (17) for $\tilde{L}_{\gamma(\xi, t)} \eta$ and (57) for P_γ , one finds successively

$$\begin{aligned} \nabla_\xi \eta &= \frac{d}{dt} \Big|_{t=0} P_{\gamma^{-1}(\xi, t)} \left(\eta + \frac{t}{2}[\xi, \eta] + O(t^2) \right) = \\ &= \frac{d}{dt} \Big|_{t=0} \eta + \frac{t}{2}([\xi, \eta] - B(\xi, \eta) - B(\eta, \xi)) + O(t^2) \quad C.Q.F.D. \end{aligned}$$

Remark. Parallel transport π_γ on a riemannian manifold preserves the scalar product $\langle a, b \rangle$. But the left-invariant product of left-invariant vector-fields $\langle a, b \rangle$ is constant. Hence ∇_e is an antisymmetric operator :

$$\langle \nabla_e a, b \rangle + \langle \nabla_e b, a \rangle = 0. \quad (59)$$

One checks easily that the expression (55) satisfies the antisymmetry relation (59).

Proof of theorem 5. Let ξ, η be two left-invariant vector-fields. Then the vector-fields $[\xi, \eta]$, $\nabla_\xi \eta$ and $\nabla_\eta \xi$ also are left-invariant. According to (54) and (55), their values at point e are

$$\begin{aligned} \nabla_\xi \eta &= \alpha - \delta, & \nabla_\eta \xi &= -(\alpha + \delta), \\ \nabla_\xi \xi &= -2B_\xi, & \nabla_\eta \eta &= -2B_\eta \end{aligned} \quad (60)$$

Substituting (60) in (49) and using (59), one finds

$$\langle -\nabla_\xi \nabla_\eta \xi, \eta \rangle = \langle \nabla_\xi \eta, \nabla_\eta \xi \rangle = -\langle \alpha - \delta, \alpha + \delta \rangle, \quad (61)$$

$$\langle \nabla_\eta \nabla_\xi \xi, \eta \rangle = \langle -\nabla_\xi \xi, \nabla_\eta \eta \rangle = -4\langle B_\xi, B_\eta \rangle. \quad (62)$$

It follows from (55) that

$$\begin{aligned} -\langle \nabla_{\{\xi, \eta\}} \xi, \eta \rangle &= \langle \nabla_{[\xi, \eta]} \xi, \eta \rangle = \frac{1}{2} \langle [[\xi, \eta], \xi], \eta \rangle - \frac{1}{2} \langle B([\xi, \eta], \xi), \eta \rangle \\ &\quad - \frac{1}{2} \langle B(\xi, [\xi, \eta]), \eta \rangle. \end{aligned} \quad (63)$$

Taking into account (10), (11), (54), one finds

$$\left. \begin{aligned} \langle [[\xi, \eta], \xi], \eta \rangle &= \langle -B(\eta, \xi), [\xi, \eta] \rangle, \\ \langle B([\xi, \eta], \xi), \eta \rangle &= \langle [\xi, \eta], [\xi, \eta] \rangle, \\ \langle B(\xi, [\xi, \eta]), \eta \rangle &= \langle -B(\xi, \eta), [\xi, \eta] \rangle. \end{aligned} \right\} \quad (64)$$

Substituting (64) in (63), one finds in accordance with (54)

$$-\langle \nabla_{\{\xi, \eta\}} \xi, \eta \rangle = -2\langle \alpha, \alpha \rangle + 2\langle \alpha, \beta \rangle. \quad (65)$$

Finally, substituting (61), (62), (65) in (49), one finds

$$C_{\xi, \eta} = -\langle \alpha - \delta, \alpha + \delta \rangle - 4\langle B_\xi, B_\eta \rangle - 2\langle \alpha, \alpha \rangle + 2\langle \alpha, \beta \rangle,$$

which is equivalent to (53). Theorem 5 is proved.

9 The case of a diffeomorphism group

Let D be a bounded domain in a Riemann space. We shall denote $\text{SDiff } D$ the group of volume preserving diffeomorphisms of the domain D . Even though $\text{SDiff } D$ is not a Lie group, it is interesting to see the assertions which the formulas of the previous paragraphs become transformed into, in this case. Such a “heuristic” study leads to propositions which can be rigorously checked afterwards (see [2]). In accordance with the heuristic character of this and the following paragraph, we leave aside differentiability problems and other similar details.

The algebra $\mathfrak{U}$ corresponding to the group $\text{SDiff } D$ is made of the vector-fields v tangent to D such that

$$\text{div } v = 0, \quad \text{and} \quad (v, n) = 0 \quad \text{on} \quad \partial D. \quad (66)$$

One defines the scalar product in the algebra $\mathfrak{U}$ by the formula

$$\langle v_1, v_2 \rangle = \int_D (v_1 \cdot v_2) dx, \quad (67)$$

where $(\cdot)$ is the scalar product which determines the metric in D , and dx is the riemannian volume element on D .

In order to write the formulas of the previous paragraphs for the group $\text{SDiff } D$, we shall use some notions from vector analysis. According to (50), the Lie commutator $[v_1, v_2]$ of two vector-fields coincide with their Poisson bracket $\{v_1, v_2\}$.

Let ω^k be a differential form of degree k on D , ξ^k a k -vector $\xi^k = \xi_1 \wedge \cdots \wedge \xi_k$. The value of the form ω^k on the polyvector ξ^k will be denoted by $\omega^k | \xi^k$. The convolution operation

$$i(\xi^k) : \omega^l \rightarrow \omega^{l-k}$$

is defined by the formula

$$i(\xi^k) \omega^l | \eta^{l-k} = \omega^l | \xi^k \wedge \eta^{l-k} \quad (68)$$

for any $(l-k)$ -vector η^{l-k} .

The riemannian structure associates with each tangent vector c a 1-form ω_c^1 , namely

$$\omega_c^1 | \xi = i(\xi) \omega_c^1 = (c, \xi) \quad (69)$$

for all tangent vectors ξ .

Theorem 6. Let a and c be vector fields in $\mathfrak{U}$. The vector-field $B(c, a) \in \mathfrak{U}$ is given by the formula (see (10)):

$$\boxed{\omega_B^1 = i(a) d\omega_c^1 + d\alpha} \quad (70)$$

where α is the univalent function on D defined by (70) and $B \in \mathfrak{U}$ (see (66)).

Proof of theorem 6. – According to H. Weyl's orthogonal decomposition theorem, each vector-field ξ on D admits a unique decomposition into the sum of two orthogonal vector-fields

$$\xi = \xi_1 + \xi_2 \quad \text{where} \quad \xi_1 \in \mathfrak{U}, \quad \xi_2 = \text{grad } \alpha,$$

α being a univalent function on M . Applying the decomposition theorem to the vector-field which corresponds to the form $i(a)d\omega_c^1$, one finds a unique decomposition into ω_b^1 and $-d\alpha$. So that formula (70) determines a unique field $B \in \mathfrak{U}$.

Formula (10) determines the scalar products $\langle B, b \rangle$ for all $b \in \mathfrak{U}$:

$$\langle B(c, a), b \rangle = \langle [a, b], c \rangle \quad (10)$$

Hence, there cannot exist more than one vector $B(c, a)$ satisfying (10). Therefore, in order to prove theorem 6 it remains to prove that the function B , determined by (70), satisfies the identity (10) for all $b \in \mathfrak{U}$.

One shall make use of the known identities (71), (76) in vector analysis

$$d(\omega^k \wedge \omega^l) = d\omega^k \omega^l + (-1)^k \omega^k \wedge d\omega^l \quad (71)$$

$$\omega^k \wedge \omega^l = (-1)^{kl} \omega^l \wedge \omega^k, \quad (72)$$

$$i(a)(\omega^k \omega^l) = (i(a)\omega^k) \wedge \omega^l + (-1)^k \omega^k \wedge (i(a)\omega^l), \quad (73)$$

$$i(a \wedge b)\omega = i(b)i(a)\omega \quad \text{for all vectors } a, b \quad (74)$$

Let τ be the riemannian volume on D (an n -form, $n = \dim D$).

Then, for all tangent vector-fields a, b , one has

$$(a, b)\tau = \omega_a^1 \wedge (i(b)\tau), \quad (75)$$

$$\boxed{d(i(a \wedge b)\tau) = i(\{a, b\})\tau + i(a)di(b)\tau - i(b)di(a)\tau.} \quad (76)$$

It follows from (67), (50) and (75) that for all vector-fields a, b, c

$$\langle [a, b], c \rangle = \int_D (\{a, b\}, c)\tau = \int_D \omega_c^1 \wedge i(\{a, b\})\tau. \quad (77)$$

For two fields $a, b \in \mathfrak{U}$, one finds according to (66)

$$di(a)\tau = \tau \operatorname{div} a = 0, \quad di(b)\tau = \tau \operatorname{div} b = 0. \quad (78)$$

According to (78), it follows from (76) that for $a, b \in \mathfrak{U}$,

$$di(a \wedge b)\tau = i(\{a, b\})\tau. \quad (79)$$

It follows from (79), (71) and Stokes formula that

$$\int_D \omega_c^1 \wedge i(a \wedge b) \tau = \int_D d\omega_c^1 \wedge i(a \wedge b) \tau - \int_{\partial D} \omega_c^1 \wedge i(a \wedge b) \tau. \quad (80)$$

According to (66), for the fields $a, b \in \mathfrak{U}$, the integral along ∂D is equal to 0. Indeed, let $\xi^{n-1} = \xi_1 \wedge \cdots \wedge \xi_{n-1}$ be a polyvector tangent to ∂D . According to (68),

$$\omega_c^1 \wedge i(a \wedge b) \tau | \xi = \sum_i \pm (\omega_c^1 | \xi_i) (\tau | a \wedge b \wedge \xi'_i)$$

where

$$\xi'_i = \xi_1 \wedge \cdots \wedge \xi_{i-1} \wedge \xi_{i+1} \wedge \cdots \wedge \xi_n.$$

But $\tau | a \wedge b \wedge \xi'_i = 0$, because the n -vector $a \wedge b \wedge \xi'_i$ is tangent to the manifold ∂D of dimension $n-1$. Hence, for $a, b \in \mathfrak{U}$

$$\int_{\partial D} \omega_c^1 \wedge i(a \wedge b) \tau = 0. \quad (81)$$

It follows from (72), (74), that

$$d\omega_c^1 \wedge i(a \wedge b) \tau = (i(a \wedge b) \tau) \wedge d\omega_c^1 = (i(b)i(a) \tau) \wedge d\omega_c^1. \quad (82)$$

Let us notice that, being of degree $n+1$, the form $(i(a) \tau) \wedge d\omega_c^1$ is equal to zero. Hence it follows from (73) that

$$(i(b)i(a) \tau) \wedge d\omega_c^1 = (-1)^n (i(a) \tau) \wedge (i(b) d\omega_c^1). \quad (83)$$

The form $\tau \wedge i(b) d\omega_c^1$ is also of degree $n+1$, hence equal to zero. Therefore (73) gives

$$(i(a) \tau) \wedge (i(b) d\omega_c^1) = (-1)^{n+1} (\tau \wedge i(a) i(b) d\omega_c^1) = (-1)^n (i(b) i(a) d\omega_c^1) \tau. \quad (84)$$

Comparing the formulas (77), (80), (81), (82), (83), (84), one finds

$$\langle [a, b], c \rangle = \int_D (i(b) i(a) d\omega_c^1) \tau. \quad (85)$$

It follows from (70) and (69), that

$$i(b) i(a) d\omega_c^1 = i(b) (\omega_B^1 - d\alpha) = (b, B - \text{grad } \alpha). \quad (86)$$

According to the orthogonal decomposition theorem, the field $b \in \mathfrak{U}$ is orthogonal to $\text{grad } \alpha$. Hence it follows from (85) and (86) :

$$\langle [a, b], c \rangle = \int_D (b, B - \text{grad } \alpha) \tau = \langle B(c, a), b \rangle \quad C.Q.F.D.$$

The tridimensional case. If $n = 3$, the formula (70) takes the form

$$\boxed{B(c, a) = \text{rot } c \wedge a + \text{grad } \alpha} \quad (87)$$

where $\wedge$ is the vector product associated to the metric¹⁰ of D . The proof of formula (87) is immediate.

The bidimensional case. Let D be a riemannian manifold of dimension 2, possibly with boundary ∂D . The vector-fields $v \in \mathfrak{U}$ admit “*stream functions*” ψ . Indeed, let $v \in \mathfrak{U}$, I the operator of right angle rotation “to the right”. Then, for each contour which is homologous to zero, $\gamma = d\sigma$,

$$0 = \iint_{\sigma} \text{div } v \, dx = \oint_{\partial \sigma} v \, dn = \oint_{\partial \sigma} Iv \, dx$$

hence the 1-form $(Iv)dx$ is closed.

So

$$\boxed{v = I \text{grad } \psi = \text{rot } \psi} \quad (88)$$

where ψ is the, possibly multivalued, “*stream function*”. it follows from (71), that the *stream function* $\psi_{\{v_1, v_2\}}$ of the commutator $\{v_1, v_2\}$ of the two fields $v_1, v_2 \in \mathfrak{U}$ is the *jacobian of the stream functions of these fields* :

$$\psi_{\{v_1, v_2\}} = J(\psi_{v_1}, \psi_{v_2}) = v_1 \wedge v_2. \quad (89)$$

Theorem 6 now becomes the following assertion: let ψ_c, ψ_a be the stream functions of the fields $c, a \in \mathfrak{U}$. Then the vector-field $B(c, a)$ is given by the formula

$$\boxed{B = -\Delta \psi_c \text{grad } \psi_a + \text{grad } \alpha,} \quad (90)$$

where $\Delta = \text{div grad} = -\text{rot rot}$

One deduces formula (90) from (87). Now $c = \text{rot } \psi_c, a = \text{rot } \psi_a, \text{rot } c \wedge a = \text{rot rot } \psi_c \cdot \text{grad } \psi_a$. In the special case of euclidean space x, y the formulas (88), (89), (90) become

$$v_x = \frac{\partial \psi}{\partial y}, v_y = -\frac{\partial \psi}{\partial x}, \psi_{\{v_1, v_2\}} = \frac{\partial \psi_{v_1}}{\partial x} \frac{\partial \psi_{v_2}}{\partial y} - \frac{\partial \psi_{v_1}}{\partial y} \frac{\partial \psi_{v_2}}{\partial x}, \Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}. \quad (91)$$

10 Applications to hydrodynamics

Let, as in §9, $SDiff D$ be the group of those diffeomorphisms of a bounded riemannian domain D which preserve the volume element.

¹⁰ Changing the orientation of D changes simultaneously the signs of rot and $\wedge$.

It follows from the principles of Mechanics that *the geodesics of the right invariant metric corresponding to (67) are the ideal fluid flows in D .*

Indeed, let us consider an ideal fluid (incompressible, inviscid, homogeneous), which fills the domain D . The transformation

$$g(t) : D \rightarrow D$$

which displaces each particle of the fluid from its position at time 0 to its position at time t preserves the volume element. Hence, $g(t) \in SDiff D$. According to the least action principle, the curve $g(t)$ is a geodesic of the group $SDiff D$, because it extremizes the integral $\int T dt$, where T is the kinetic energy:

$$T = \frac{1}{2} \int_D v^2 dx$$

where v is the velocity field¹¹. Now

$$g(t+dt) = (\exp(vdt))g(t).$$

Hence the velocity field is the right translation of the vector $\dot{g}$ in the algebra $\mathfrak{L}$:

$$v = \tilde{R}_{g^{-1}} \dot{g}.$$

Hence, $g(t)$ is the geodesic of a right invariant metric, corresponding to the form (67).

The formulas of the preceding paragraphs correspond to left invariant metrics. In order to obtain the corresponding formulas in the right invariant case, it is enough, by (52), to change the signs in front of all the commutators and the B operations; this does not change the quadratic forms (44), (45), (53).

In accordance with (32) and (52), the velocity field of the ideal fluid satisfies the equation

$$\frac{\partial v}{\partial t} = -B(v, v). \quad (92)$$

Comparing to (87), one finds the *equation of motion "in the form of Bernoulli"* for the ideal fluid in a riemannian domain D of dimension 1:

$$\frac{\partial v}{\partial t} = v \wedge \text{rot } v + \text{grad } \alpha, \quad \text{div } v = 0, \quad (v, n) = 0 \text{ on } \partial D. \quad (93)$$

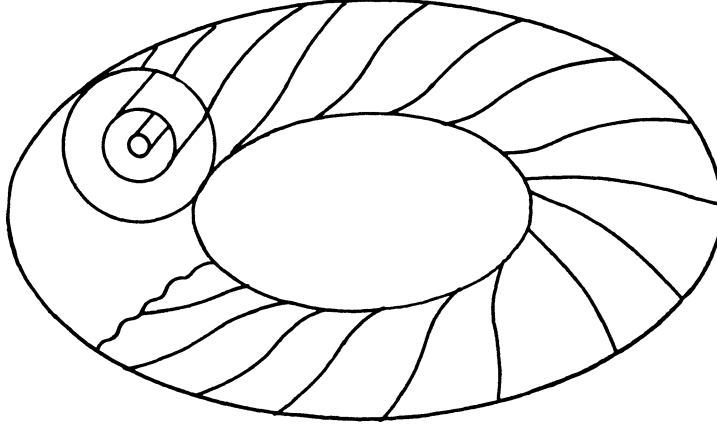
In the case of a steady flow, (93) and (76) imply

$$v \wedge \text{rot } v = -\text{grad } \alpha, \quad \{v, \text{rot } v\} = 0. \quad (94)$$

¹¹ From the mathematician's point of view, one can consider this principle as a definition of the ideal fluids. See [14, 11].

Thus, *the velocity field of a steady flow commutes with its curl*. One immediately deduces from this a topological classification of analytic steady flows of ideal fluids in three-dimensional domains [4].

In particular, the domain D turns out to be divided by an analytic subspace into a finite number of open “cells”, fibered by tori or cylinders generated by the stream lines¹². The stream lines are closed on the cylinders, closed or dense on the tori (fig. 2, 3).



More precisely, the following theorem holds true

Theorem 7. (see [4]). *Let us suppose that*

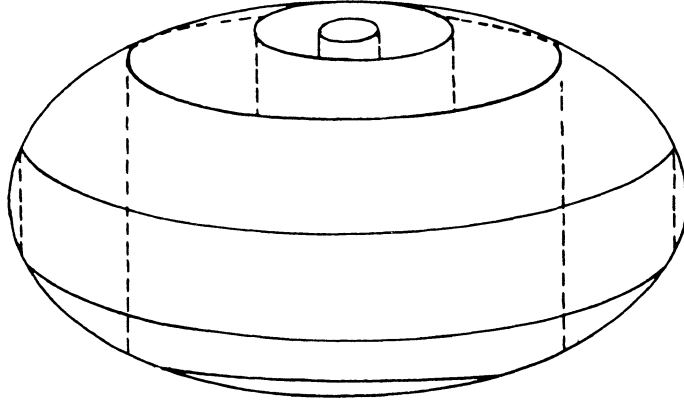
- 1) *The three-dimensional domain D is connected and compact.*
- 2) *The steady flow velocity field v , the domain D and the boundary ∂D are real analytic.*
- 3) *the velocity and the curl are not everywhere collinear:*

$$v \wedge \text{rot } v \neq 0.$$

Then almost every stream line is either closed or everywhere dense on analytic tori analytically embedded in D : all the other stream lines make up a true compact analytic subset in D .

Remark. The analyticity condition is probably not very important. The non collinearity condition is probably important.

¹² The trajectories of the velocity field v .



Because the flows with $\text{rot } v = \lambda v^{13}$ allow probably for stream lines with a topology as complicated as the one of orbits in celestial mechanics (see [5], fig. 6). In the same way, the stream lines of stationary flows of viscous fluids may possess a complicated topology.

Proof of theorem 7 According to (94), the stream lines, as the curl lines, are orthogonal to $\text{grad } \alpha$. Hence they belong to the level surfaces $\alpha = \text{constant}$. Let us study the topology of these surfaces.

We shall call α_0 a critical value if there exists a point x in D , where $\text{grad } \alpha(x) = 0$ and $\alpha(x) = \alpha_0$, or if there exists a point x in ∂D , where $\text{grad } \alpha(x)$ is orthogonal to ∂D and $\alpha(x) = \alpha_0$. From the analyticity of α and ∂D , it follows that there is only a finite number of critical values. As the function α is not constant (non collinearity), the points x such that $\alpha(x)$ is critical make up a finite number of analytic surfaces and curves in D .

These surfaces divide $D \cup \partial D$ into “cells”, in each of which $\text{grad } \alpha \neq 0$ and $\text{grad } \alpha$ is not orthogonal to ∂D . We shall show that in each cell, the stream lines are closed or everywhere dense on tori.

Let us consider a connected component C of a non-critical level surface $\alpha(x) = \alpha_1$, the value α_1 being non-critical. This component C , either

- 1) does not intersect ∂D , then C is a closed analytic surface in D , or
- 2) intersects ∂D , $C \cap \partial D = \partial C$ is made of a finite number of closed analytical curves (the intersection is transversal because $\text{grad } \alpha$ is not orthogonal to ∂D).

If C is a surface without boundary, the group $\mathbb{R}^2$ acts on C globally and without fixed point. Indeed, let $(s, t) \in \mathbb{R}^2, x \in C$. Consider the solution $y(s_0; s, t; \tau)$ of the system

$$\frac{dy}{d\tau} = sr + tv, y(x_0; s, t; 0) = x_0, (r = \text{rot } v).$$

¹³ Example: $D = \mathbb{T}^3, v_x = A \sin z + C \cos y, v_y = B \sin x + C \cos z, v_z = C \sin y + B \cos x$.

As the vector-fields r and v commute (94), the mapping

$$x : C \times \mathbb{R}^2 \rightarrow C$$

given by $x(x_0; s, t) = y(x_0; s, t; 1)$, defines the action of $\mathbb{R}^2$ on C .

But α_1 is non-critical, hence $v \wedge r \neq 0$ everywhere on C , so that $\mathbb{R}^2$ acts without fixed point. Hence C is a torus, and the stream line $x(x_0; 0, t)$ is the orbit of a one parameter subgroup; this proves theorem 7 when $\partial C = 0$.

Suppose now that $\partial C \neq 0$. Then ∂C is composed of a finite number of closed stream lines in ∂D . Let $x(x_0; 0, \tau)$ be one of these components of ∂C :

$$x(x_0; 0, \tau + T) = x(x_0; 0, \tau) \quad \text{for all } \tau.$$

Due connectedness of C and the condition of non-collinearity, each point of C can be written in the form

$$x_1 = x(x_0; s, t).$$

Hence the commutativity of $\mathbb{R}^2$ implies

$$x(x_1; 0, T) = x_1$$

that is, the stream line through x_1 is closed. One sees also that C has the topology of the cylinder $S^1 \times I^1$. Theorem 7 is proved.

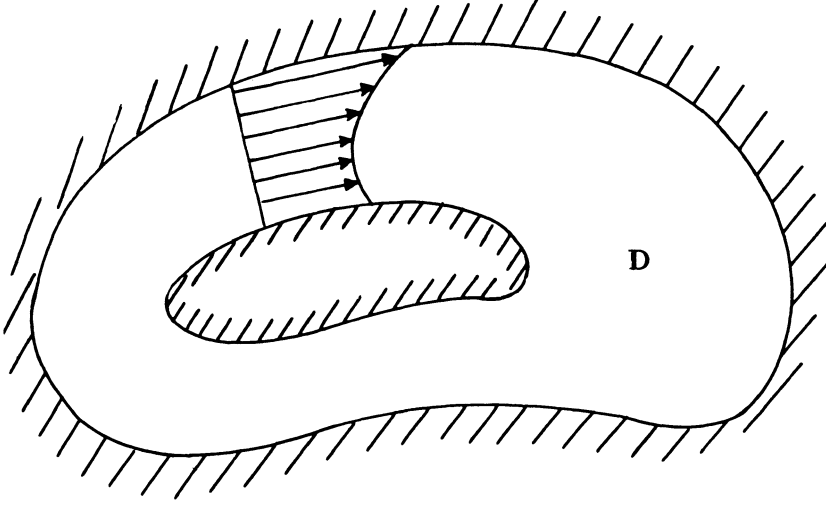
In hydrodynamics, the variational equation (46) in the neighborhood of the steady flow (94) corresponds to the Rayleigh equation (Orr-Sommerfeld equation for a non viscous fluid). The quadratic form (44) is a first integral of this linear equation. In accordance with theorem 4, it is sufficient for the stability of the steady flow that the quadratic form (44) be definite (positive or negative).

It seems that, for three-dimensional flows, the form (44) is never definite. It is only for two-dimensional flows that theorem 4 implies effective stability criteria.

Let, for example, D be a planar annulus (fig. 4).

The steady flow in D is stable if its stream function satisfies the concavity condition of its velocity profile.

$$\frac{\nabla \psi}{\nabla \Delta \psi} > 0. \quad (95)$$



Indeed

Theorem 8. *The quadratic form (44) is*

$$2\delta^2 E = \iint_D (\delta v)^2 + \frac{\nabla \psi}{\nabla \Delta \psi} (\delta r)^2 dx dy \quad (96)$$

where $\delta r = \text{rot } v$.

Remark. In (95) and (96) the ratio of the vectors $\nabla \psi$ and $\nabla \Delta \psi$ is a scalar, because for the steady flow these vectors are collinear (see (94) and (91)).

The condition (95) implies: the form (96) is definite positive. Hence, according to theorem 4, the steady flow (95) is stable. One will find a rigorous proof of stability in [2].

Proof of theorem 8. According to formulæ (41), (44), (67), the quadratic form (44) is

$$2\delta^2 E = \iint_D \delta v^2 + \delta v \{f, v\} dx dy, \text{ where } \delta v = B(v, f) = r \wedge f + \text{grad } \alpha, \quad (97)$$

$$r = -\Delta \psi.$$

Integrating by parts, one finds according to (76) and (71)

$$\iint_D \delta v \{f, v\} dx dy = \iint_D \delta v \text{rot } (f \wedge v) dx dy = \iint_D (\delta r) (f \wedge v) dx dy, \quad (98)$$

with obvious notations.

Formula (88) implies

$$f \wedge v = f \wedge I \nabla \psi = -(f, \nabla \psi). \quad (99)$$

On the other hand, it follows from (97), (51) and (91), that

$$\begin{aligned} \delta r = \{r, f\} &= -\Delta \psi_x f_1 - \Delta \psi f_{1_x} - \Delta \psi_y f_2 - \Delta \psi f_{2_y} \\ &= -\Delta \psi (f_{1_x} + f_{2_y}) - \Delta \psi_x f_1 - \Delta \psi_y f_2 = -(f, \nabla \Delta \psi). \end{aligned} \quad (100)$$

Comparing formulæ (99) and (100), one finds

$$f \wedge v = \frac{\nabla \psi}{\nabla \Delta \psi} \delta r. \quad (101)$$

Formula (96) results directly from (97), (98) and (101). Theorem 8 is proved.

Remark. The steady flow may be stable even in case formula (96) is neither positive definite, nor negative definite.

Indeed, the circular flow in a circular annulus

$$(\psi = \psi(x^2 + y^2))$$

is stable, if $\frac{\nabla \psi}{\nabla \Delta \psi}$ does not change sign. For the proofs see [1]; one uses theorem 4 and the first integral of angular momentum.

In the same way, the planar-parallel flow

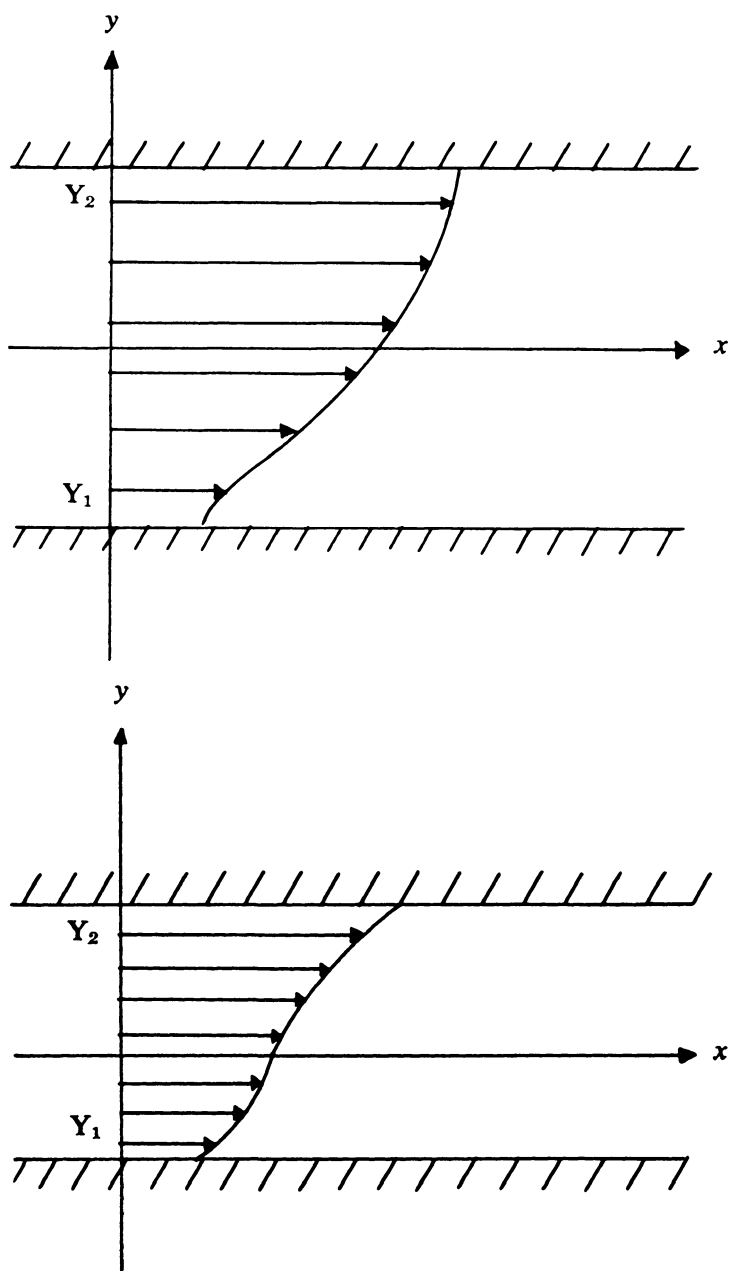
$$\vec{v} = v(y), 0; \quad Y_1 \leq y \leq Y_2; \quad x \bmod X$$

is stable, if $v_{yy} \neq 0$, that is if the velocity profile has no inflexion points (non linear analogue of Rayleigh's theorem, see [12, 9]). Moreover, a flow with a single inflexion point is stable in the case of figure 5 ($v/v_{yy} > 0$), while in the case of figure 6 ($v/\frac{1}{2}u_{yy} < 0$), it is stable in a sufficiently narrow band. For example, if

$$v(y) = \sin y, Y_1 + Y_2 = 0, X = 2\pi,$$

the flow is stable for $|Y_2 - Y_1| > \pi$. For the proofs, see [1] and [2]; one uses theorem 4 and the first integral of linear momentum.

The methods of [1, 2] also apply to the other two-dimensional motions. For example, in [7], L.A. Dikii has studied spherical flows.



The problem: does there exist a stable flow among the three-dimensional steady flows of an ideal fluid ? remains open.

11 The riemannian curvature of diffeomorphisms groups

The expression (53) for the curvature of the Lie group endowed with a left invariant metric¹⁴, is still meaningful in the case of the infinite dimensional group $SDiffD$ of diffeomorphisms of a riemannian domain D . I call this same expression the “riemannian curvature” of the group $SDiffD$.

As an example, in this paragraph we shall compute the curvature of $SDiff\mathbb{T}^2$, the diffeomorphisms of a torus $\mathbb{T}^2$, endowed with a locally euclidean metric. The computations show that, in “good enough” sections, the curvature is *finite*. One finds many sections in which the curvature is *negative* (theorem 9). But one knows since Hadamard [10], that negative curvature implies the instability of geodesics. As the geodesics of the group $SDiffD$ are ideal fluid flows, this shows the instability¹⁵ of some flows.

On the other hand, one finds sections in which the curvature is *positive*. I do not know if there exists conjugate points on the group $SDiff\mathbb{T}^2$.

Let us first prove some auxiliary formulæ. As we saw in §9, the algebra $\mathfrak{U}$ of the group $SDiff\mathbb{T}^2$ may be considered as the algebra of stream functions $\psi(x, y)$. From now on we shall restrict ourselves to the subalgebra $\mathfrak{U}_0$ of *univalent* stream functions¹⁶.

Let $\mathbb{C}\mathfrak{U}_0$ be the complex extension of the algebra $\mathfrak{U}_0$. The extension of the operations $[\cdot, \cdot], \langle \cdot, \cdot \rangle, B(\cdot, \cdot), \nabla, R$, defined in the former paragraphs, is immediate; one has

$$\begin{aligned} [\lambda z, \lambda' z'] &= \lambda \lambda' [z, z']; \quad \langle \lambda z, \lambda' z' \rangle = \lambda \lambda' \langle z, z' \rangle, \\ B(\lambda z, \lambda' z') &= \lambda \lambda' B(z, z'), \\ \nabla_{\lambda z} \lambda' z' &= \lambda \lambda' \nabla_z z'; \end{aligned} \quad (102)$$

$$\langle R(\lambda z, \lambda' z') \lambda'' z'', \lambda''' z''' \rangle = \lambda \lambda' \lambda'' \lambda''' \langle R(z, z') z'', z''' \rangle, \quad (103)$$

for all $z \in \mathbb{C}\mathfrak{U}_0$ and all $\lambda \in \mathbb{C}$.

Obviously, the operations extended in this way satisfy the algebraic identities (6), (10), (11), (45), (53), (59). The Riemann tensor appearing in (103) is defined in the real case by

$$R(X, Y)Z = -\nabla_X \nabla_Y Z + \nabla_Y \nabla_X Z + \nabla_{[X, Y]} Z. \quad (104)$$

¹⁴ Of course, in the case of a right invariant metric, the curvature of a corresponding section is the same.

¹⁵ here, the word instability has a meaning which differs from the one in par. 10: the question is about the proximity of the curves $g(t)$ in the group and not the the one of their images $v(t)$ in the algebra.

¹⁶ This subalgebra is an ideal. The associated invariant subgroup (group of motions preserving the “center of mass”) is a totally geodesic submanifold of $SDiff\mathbb{T}^2$, as results from the law of conservation of linear momentum. Hence, the curvature of the corresponding sections in the group $SDiff\mathbb{T}^2$ and in the subgroup are the same.

It is related to the the curvature (49) in the section defined by the two orthonormal vectors ξ, η , by the formula

$$C_{\xi\eta} = \langle R(\xi, \eta)\xi, \eta \rangle, \quad (105)$$

and it satisfies the symetry and Bianchi identities:

$$\langle R(X, Y)Z, W \rangle = -\langle R(X, Y)W, Z \rangle = \langle R(Z, W)X, Y \rangle \quad (106)$$

$$\langle R(X, Y)Z, W \rangle + \langle R(X, Z)W, Y \rangle + \langle R(X, W)Y, Z \rangle = 0. \quad (107)$$

The relations (106) and (107) remain valid in $\mathbb{C}\mathfrak{U}_0$.

Now, set

$$e_k = e^{i(k, x)},$$

a univalent function on the torus $\mathbb{T}^2 = \mathbb{R}^2 / \Gamma$, $\mathbb{R}^2 = \{x\}$, k a vector in the euclidean plane $\mathbb{R}^2$. Obviously, the admissible vectors k form a lattice and the functions e_k , $k \neq 0$, form a basis in the space $\mathbb{C}\mathfrak{U}_0$. One shall compute the operations (102), (103) on the basis vectors.

Let S be the area of the torus $\mathbb{T}^2$, and $R_{k,l,m,n} = \langle R(e_k, e_l)e_m, e_n \rangle$.

Lemma 10. *The following formulae hold true:*

$$\langle e_k, e_l \rangle = 0 \quad \text{for } k+l \neq 0; \quad \langle e_k, e_{-k} \rangle = k^2 S; \quad (108)$$

$$[e_k, e_l] = (k \wedge l) e_{k+l}; \quad (109)$$

$$B(e_k, e_l) = b_{k,l} e_{k+l}, \quad b_{k,l} = (k \wedge l) \frac{k^2}{(k+l)^2}; \quad (110)$$

$$\nabla_{e_k} e_l = d_{l,k+l} e_{k+l}, \quad d_{u,v} = \frac{(v \wedge u)(u \cdot v)}{v^2} \quad (111)$$

$$\boxed{R_{k,l,m,n} = (a_{l,n} a_{k,m} - a_{l,m} a_{k,n}) S,} \quad (112)$$

for $k+l+m+n=0$ one has

$$\boxed{a_{u,v} = \frac{(u \wedge v)^2}{|u+v|};} \quad (113)$$

if $k+l+m+n \neq 0$, then $R_{k,l,m,n} = 0$.

Proof of lemma 10. Formula (108) is an immediate consequence of definition (67), and formula (109) is a consequence of formulæ(51) and (52).

Then, it follows from formulæ(10) and (109) that

$$\langle B(e_k, e_l), e_m \rangle = (l \wedge m) \langle e_{l+m}, e_k \rangle. \quad (114)$$

According to (108), formula (114) shows that $B(e_k, e_l)$ is orthogonal to e_m for $k + l + m \neq 0$. Hence $B(e_k, e_l) = b_{k,l}e_{k+l}$. One finds the expression (110) for $b_{k,l}$ by using (114) and (108) with $m = -k - l$.

According to (109), (110), it follows from (55) that

$$\nabla_{e_k} e_l = \frac{1}{2}(k \wedge l) \left(1 - \frac{k^2 - l^2}{(k+l)^2} \right) e_{k+l}. \quad (115)$$

But obviously

$$\frac{1}{2}(k \wedge l) \left(1 - \frac{k^2 - l^2}{(k+l)^2} \right) = \frac{\langle l, k+l \rangle}{(k+l)^2}. \quad (116)$$

Formulae (115) and (116) imply (111).

Then, substituting (111) into (104), one finds according to (103)

$$\nabla_{e_k} \nabla_{e_l} e_m = d_{l+m, k+l+m} d_{m, l+m} e_{k+l+m}, \quad (117)$$

$$\nabla_{e_l} \nabla_{e_k} e_m = d_{k+m, k+l+m} d_{m, k+m} e_{k+l+m}, \quad (118)$$

$$\nabla_{[e_k, e_l]} e_m = (k \wedge l) d_{m, k+l+m} e_{k+l+m}. \quad (119)$$

Comparing formulae (104), (117), (118), (119) and (108), one finds $R_{k,l,m,n} = 0$ for $k + l + m + n \neq 0$, and for $k + l + m + n = 0$, one finds

$$R_{k,l,m,n} = (d_{k+m,n} d_{m,k+m} - d_{l+m,n} d_{m,l+m} + (k \wedge l) d_{m,n}) n^2 S, \quad (120)$$

(one has used the fact that, according to (111), $d_{u,v}$ is symmetric in u and in v)).

Now, let $k + l + m + n = 0$. According to (111), one has

$$\left. \begin{aligned} -d_{k+m,n} &= \frac{l \wedge (k+m)}{n^2} (k+m, n); \\ -d_{m,l+m} &= \frac{m \wedge l}{(l+m)^2} (l+m, m); \\ -d_{m,k+m} &= \frac{m \wedge k}{(k+m)^2} (k+m, m); \\ -d_{m,n} &= \frac{(l+k) \wedge m}{n^2} (m, n). \\ -d_{l+m,n} &= \frac{k \wedge (l+m)}{n^2} (l+m, n); \end{aligned} \right\} \quad (121)$$

It follows from (121) and (120) that:

$$S^{-1} R_{k,l,m,n} = A(l \wedge k)(m \wedge k) + (A - B)(l \wedge m)(m \wedge k) + B(k \wedge l)(l \wedge m), \quad (122)$$

where

$$\begin{aligned} A &= \frac{(k+m, m)(k+m, n)}{(k+m)^2} - (m, n); \\ B &= \frac{(l+m, m)(l+m, n)}{(l+m)^2} - (m, n); \end{aligned} \quad (123)$$

Now one shall use the identity

$$\frac{(a, b)(a, c)}{a^2} + \frac{(a \wedge b)(a \wedge c)}{a^2} = (b, c) \quad (124)$$

where a, b, c are three arbitrary vectors in the euclidean plane.

(Formula (124) is best known in the form

$$\cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos(\alpha - \beta).$$

It follows from (123), (124) and $k + l + m + n = 0$, that

$$A = \frac{(k \wedge m)(l \wedge n)}{(k+m)^2}, \quad B = \frac{(l \wedge m)(k \wedge n)}{(l+m)^2}. \quad (125)$$

Formula (122) is equivalent to

$$S^{-1}R_{k,l,m,n} = A(m \wedge k)(l \wedge (k+m)) + B(l \wedge m)(k \wedge (l+m)). \quad (126)$$

Taking into account that $k + l + m + n = 0$, (126) reduces to

$$S^{-1}R_{k,l,m,n} = A(m \wedge k)(n \wedge l) + B(l \wedge m)(n \wedge k). \quad (127)$$

Comparing (127) and (125), one finds

$$S^{-1}R_{k,l,m,n} = \frac{(k \wedge m)^2(l \wedge n)^2}{(k+m)^2} - \frac{(l \wedge m)^2(k \wedge n)^2}{(l+m)^2}. \quad (128)$$

Formula (128) is equivalent to (112), (113), because $k + m = -(l + n)$ and $l + m = -(k + n)$. This proves lemma 10.

Theorem 9. *In each section which contains the vector $\xi \in \mathfrak{L}$, corresponding to the stream function $\cos kx$, the curvature of the group $SDiff \mathbb{T}^2$ is non-negative.*

Theorem 9 is an immediate consequence of formula (129) below.

Lemma 11. *Let*

$$2\xi = e_k + e_{-k}, \quad \eta = \sum x_l e_l$$

be two real vectors in $\mathfrak{L}$, that is such that $x_{-l} = \bar{x}_l$. Then, with the notations of (113),

$$\boxed{\langle R(\xi, \eta)\xi, \eta \rangle = -\frac{S}{4} \sum_l a_{k,l}^2 |x_l + x_{l+2k}|^2} \quad (129)$$

We note a corollary of formula (129):

Corollary 1. *In the section defined by the stream functions $\xi = \cos kx, \eta = \cos lx$, the curvature of the group $S\text{Diff}\mathbb{T}^2$ is*

$$C_{\xi,\eta} = -\frac{k^2 + l^2}{4S} \sin^2 \alpha \sin^2 \beta \quad (130)$$

where S is the area of the torus, α the angle between k and l , β the angle between $k+l$ and $k-l$.

For example, for $\mathbb{T}^2 = \{x, y \bmod 2\pi\}$ the curvature in the section determined by the vector-fields $(\sin y, 0)$ and $(0, \sin x)$ is

$$C_{\xi,\eta} = -\frac{1}{8\pi^2}. \quad (131)$$

Formula (130) implies (131); here $k^2 = l^2 = 1$, $S = 4\pi^2$, $\alpha = \beta = \frac{\pi}{2}$.

Proof of formula (130). As the vectors ξ, η are orthogonal in $\mathfrak{U}$

$$C_{\xi,\eta} = -\frac{\langle R(\xi, \eta)\xi, \eta \rangle}{\langle \xi, \xi \rangle \langle \eta, \eta \rangle}. \quad (132)$$

According to (108),

$$\langle \xi, \xi \rangle = \frac{1}{2}k^2 S, \quad \langle \eta, \eta \rangle = \frac{1}{2}l^2 S. \quad (133)$$

Setting in (129), $x_l = x_{-l} = \frac{1}{2}$, one finds

$$\langle R(\xi, \eta)\xi, \eta \rangle = -\frac{S}{16}(a_{k,l}^2) + a_{k,l-2k}^2 + a_{k,-l}^2 + a_{k,-l-2k}^2. \quad (134)$$

In accordance with formula (113), $a_{k,l-2k}^2 = a_{k,-l}^2$ and $a_{k,-l-2k}^2 = a_{k,l}^2$. Hence (134) becomes

$$\langle R(\xi, \eta)\xi, \eta \rangle = -\frac{S}{8}(a_{k,l}^2 + a_{k,-l}^2). \quad (135)$$

It follows from (113) that

$$a_{k,l}^2 + a_{k,-l}^2 = (k \wedge l)^4 \left(\frac{1}{h_+^2} - \frac{1}{h_-^2} \right), \quad h_+ = k+l, \quad h_- = k-l. \quad (136)$$

One shall use the obvious identities

$$h_+^2 + h_-^2 = 2(k^2 + l^2), \quad h_+ \wedge h_- = -2(k \wedge l). \quad (137)$$

It follows from (136) and (137) that

$$a_{k,l}^2 + a_{k,-l}^2 = \frac{(k \wedge l)^2 (h_+ \wedge h_-)^2}{2h_+^2 h_-^2} (k^2 + l^2). \quad (138)$$

Comparing formulæ (138), (135), (133), (132) and taking into account that

$$(k \wedge l)^2 = k^2 l^2 \sin^2 \alpha, \quad (h_+ \wedge h_-)^2 = h_+^2 h_-^2 \sin^2 \beta,$$

one gets (130). The corollary is proved.

Proof of lemma 11. According to (103),

$$\begin{aligned} \langle R(\xi, \eta) \xi, \eta \rangle = \frac{1}{4} \sum_l (R_{k,l,k,-2k-l} x_l x_{-2k-l} + R_{-k,l,-k,2k-l} x_l x_{2k-l} \\ + R_{k,l,-k,-l} x_l x_{-l} + R_{-k,l,k,-l} x_l x_{-l}). \end{aligned} \quad (139)$$

Using (112), one finds for the coefficients of the quadratic form (139) the values

$$R_{k,l,k,-2k-l} = R_{-k,l,k,-l} = -a_{k,l}^2 S, \quad R_{k,l,-k,-l} = R_{-k,l,-k,2k-l} = -a_{k,-l}^2 S.$$

Hence, the form (139) becomes

$$\langle R(\xi, \eta) \xi, \eta \rangle = -\frac{S}{4} \sum_l [a_{k,l}^2 (x_l x_{-2k-l} + x_l x_{-l}) + a_{k,-l}^2 (x_l x_{2k-l} + x_l x_{-l})]. \quad (140)$$

But it follows from formula (113) that

$$a_{k,-l}^2 = a_{k,l-2k}^2. \quad (141)$$

Using (141), one shows easily that

$$\sum_l a_{k,-l}^2 (x_l x_{2k-l} + x_l x_{-l}) = \sum_l a_{k,l}^2 (x_{l+2k} x_{-l} + x_{l+2k} x_{-l-2k}). \quad (142)$$

It follows from (140), (142) and the reality relation $x_{-j} = \bar{x}_j$, that

$$\langle R(\xi, \eta) \xi, \eta \rangle = -\frac{S}{4} \sum_l a_{k,l}^2 (x_l \bar{x}_{l+2k} + x_l \bar{x}_l + x_{l+2k} \bar{x}_l + x_{l+2k} \bar{x}_{l+2k}),$$

which is equivalent to (129). Lemma 11 is proved.

Remark. There also exist sections in which the curvature of the group $S\text{Diff}\mathbb{T}^2$ is positive. Indeed, let us consider a section, defined by two stream lines

$$\begin{aligned} \xi &= \cos(3px - y) + \cos(3px + 2y), \\ \eta &= \cos(px + y) + \cos(px - 2y), \end{aligned} \quad (143)$$

on the torus $\mathbb{T}^2 = \{x, y \bmod 2\pi\}$. The sectional curvature is

$$C_{\xi, \eta} = \frac{\langle R(\xi, \eta)\xi, \eta \rangle}{\langle \xi, \xi \rangle \langle \eta, \eta \rangle} \rightarrow \frac{9}{8\pi^2} \quad \text{when } p \rightarrow \infty. \quad (144)$$

Proof of formula (144). Let k, l, m, n be integer planar vectors, $k + m = l + n$, and

$$\begin{aligned} \xi &= x_k \cos kx + x_m \cos mx, \\ \eta &= y_l \cos lx + y_n \cos nx. \end{aligned} \quad (145)$$

Then, according to (103), (106), (107), one finds

$$\begin{aligned} \langle R(\xi, \eta)\xi, \eta \rangle &= \frac{S}{16} (\alpha_{k,l} x_k^2 x_l^2 + \alpha_{k,n} x_k^2 y_n^2 + \alpha_{m,l} x_m^2 y_l^2 \\ &\quad + \alpha_{m,n} x_m^2 y_n^2 + \alpha x_k x_m y_l y_n), \end{aligned} \quad (146)$$

where, in accordance with formula (112),

$$\left. \begin{aligned} \alpha_{k,l} &= -2(a_{k,l}^2 + a_{k,-l}^2); & \alpha_{k,n} &= -2(a_{k,n}^2 + a_{k,-n}^2); \\ \alpha_{m,l} &= -2(a_{m,l}^2 + a_{m,-l}^2); & \alpha_{m,n} &= -2(a_{m,n}^2 + a_{m,-n}^2); \\ \alpha &= 4(2a_{k,m}a_{l,n} - a_{k,-n}a_{l,-m} - a_{k,-l}a_{m,-n}). \end{aligned} \right\} \quad (147)$$

The vectors (143) have the form (145) with $x_k = x_m = y_l = y_n = 1$, and

$$\left. \begin{aligned} k &= -3p, 1; & m &= 3p, 2; \\ l &= p, 1; & n &= -p, 2. \end{aligned} \right\} \quad (148)$$

It follows from expressions (148) that for $p \rightarrow \infty$, all the products $k \wedge l \dots$ are of order p , with the sums and differences $|k \pm l| \dots$, except

$$|k + m| = |l + n| = 3.$$

Hence, according to (113), all the $a_{k,l} \dots$ in (147) are of order p , except

$$a_{k,m} = 27p^2, \quad a_{l,n} = 3p^2.$$

Thus, one obtains for (146) the asymptotic expression:

$$\langle R(\xi, \eta)\xi, \eta \rangle \sim \frac{S\alpha}{16} \sim \frac{S \cdot 81p^4}{2}, \quad (p \rightarrow \infty). \quad (149)$$

According to (108), $\langle \xi, \xi \rangle = \frac{1}{2}(k^2 + m^2)S$; $\langle \eta, \eta \rangle = \frac{1}{2}(l^2 + n^2)S$. Hence it follows from (148) that

$$\langle \xi, \xi \rangle \sim 9p^2S, \quad \langle \eta, \eta \rangle \sim p^2S \quad (p \rightarrow \infty). \quad (150)$$

Given the area $S = 4\pi^2$ of the torus, formulæ(149) and (150) give (144). Q.E.D.

A more precise computation shows that $C_{\xi,\eta} > 0$ already for $p = 2$, that is in the section

$$\xi = \cos(6x - y) + \cos(6x + 2y), \quad \eta = \cos(2x + y) + \cos(2x - 2y).$$

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